ROCKET SPACE SHOOTER GAME DEVELOPMENT USING PYTHON

AIM

To design and develop an interactive 2D Rocket Space Shooter game using Python Turtle that allows players to control a rocket using keyboard arrow keys, shoot bullets at enemies, increase their score, and avoid collisions with enemies. The game includes a graphical interface, score tracking, lives management, game-over detection, and restart functionality.

PROCEDURE

1. Start the program and import the required Python libraries: turtle, random, and math.

2. Create the game window using Turtle and set the title as “Rocket Space Shooter”.

3. Set the background colour to black and configure the window dimensions to 800 × 600 pixels.

4. Initialize the score to zero, lives to three, and create lists to store bullets and enemies.

5. Create the rocket using a cyan triangular shape and place it near the bottom of the game window.

6. Create a text display to show the player's score and remaining lives.

7. Implement keyboard controls using the arrow keys to move the rocket in four directions: Up, Down, Left, and Right.

8. Create a shooting function that generates yellow bullets when the Spacebar is pressed.

9. Generate red circular enemies at random horizontal positions near the top of the game window.

10. Move the bullets upward and enemies downward continuously using the game loop.

11. Detect collisions between bullets and enemies. When a bullet hits an enemy, remove both objects and increase the score by 10 points.

12. Implement collision detection to identify when an enemy reaches the bottom of the screen or touches the rocket. Reduce the player's lives when either event occurs.

13. Display the Game Over message, final score, and restart instruction when the player's lives reach zero.

14. Provide a restart option using the R key to reset the score, lives, rocket position, bullets, and enemies.

15. Execute the game and verify the rocket movement, shooting, scoring, collision detection, lives management, and restart functionality.

PROGRAM CODE

```
import turtle
import random
import math

# Screen setup
screen = turtle.Screen()
screen.title("Rocket Space Shooter")
screen.bgcolor("black")
screen.setup(width=800, height=600)
screen.tracer(0)

# Game variables
score = 0
lives = 3
game_over = False
bullets = []
enemies = []

# Rocket
rocket = turtle.Turtle()
rocket.shape("triangle")
rocket.color("cyan")
rocket.penup()
rocket.setheading(90)
rocket.goto(0, -230)

# Score display
pen = turtle.Turtle()
pen.hideturtle()
pen.color("white")
pen.penup()
pen.goto(-370, 260)

# Message display
message = turtle.Turtle()
message.hideturtle()
message.color("white")
message.penup()

def update_score():
    pen.clear()
    pen.goto(-370, 260)
    pen.write("Score: " + str(score) + "   Lives: " + str(lives),
              font=("Arial", 16, "bold"))

def move_left():
    if not game_over and rocket.xcor() > -370:
        rocket.setx(rocket.xcor() - 25)

def move_right():
    if not game_over and rocket.xcor() < 370:
        rocket.setx(rocket.xcor() + 25)

def move_up():
    if not game_over and rocket.ycor() < 0:
        rocket.sety(rocket.ycor() + 20)

def move_down():
```

PROGRAM CODE (CONTINUED)

```
    if not game_over and rocket.ycor() > -270:
        rocket.sety(rocket.ycor() - 20)

def shoot():
    if not game_over:
        bullet = turtle.Turtle()
        bullet.shape("triangle")
        bullet.color("yellow")
        bullet.penup()
        bullet.setheading(90)
        bullet.shapesize(0.5, 0.5)
        bullet.goto(rocket.xcor(), rocket.ycor() + 20)
        bullets.append(bullet)

def create_enemy():
    enemy = turtle.Turtle()
    enemy.shape("circle")
    enemy.color("red")
    enemy.penup()
    enemy.goto(random.randint(-360, 360), 280)
    enemies.append(enemy)

def end_game():
    global game_over
    game_over = True
    message.clear()
    message.goto(0, 30)
    message.write("GAME OVER", align="center",
                  font=("Arial", 30, "bold"))
    message.goto(0, -20)
    message.write("Final Score: " + str(score), align="center",
                  font=("Arial", 18, "normal"))
    message.goto(0, -60)
    message.write("Press R to Restart", align="center",
                  font=("Arial", 16, "normal"))

def restart():
    global score, lives, game_over

    if not game_over:
        return

    for item in bullets + enemies:
        item.hideturtle()

    bullets.clear()
    enemies.clear()

    score = 0
    lives = 3
    game_over = False

    rocket.goto(0, -230)
    message.clear()
    update_score()
    game_loop()

def game_loop():
```

PROGRAM CODE (CONTINUED)

```
    global score, lives

    if game_over:
        screen.update()
        return

    # Move bullets
    for bullet in bullets[:]:
        bullet.sety(bullet.ycor() + 15)

        if bullet.ycor() > 300:
            bullet.hideturtle()
            bullets.remove(bullet)

    # Move enemies
    for enemy in enemies[:]:
        enemy.sety(enemy.ycor() - 3)

        if enemy.ycor() < -290:
            enemy.hideturtle()
            enemies.remove(enemy)
            lives -= 1
            update_score()

            if lives <= 0:
                end_game()
                screen.update()
                return

        elif enemy.distance(rocket) < 30:
            enemy.hideturtle()
            enemies.remove(enemy)
            lives -= 1
            update_score()

            if lives <= 0:
                end_game()
                screen.update()
                return

    # Bullet hits enemy
    for bullet in bullets[:]:
        for enemy in enemies[:]:
            if bullet.distance(enemy) < 20:
                bullet.hideturtle()
                enemy.hideturtle()

                if bullet in bullets:
                    bullets.remove(bullet)
                if enemy in enemies:
                    enemies.remove(enemy)

                score += 10
                update_score()
                break

    if random.randint(1, 25) == 1 and len(enemies) < 8:
        create_enemy()
```

PROGRAM CODE (CONTINUED)

```
    screen.update()
    screen.ontimer(game_loop, 30)

screen.listen()
screen.onkeypress(move_left, "Left")
screen.onkeypress(move_right, "Right")
screen.onkeypress(move_up, "Up")
screen.onkeypress(move_down, "Down")
screen.onkeypress(shoot, "space")
screen.onkeypress(restart, "r")
screen.onkeypress(restart, "R")

update_score()
game_loop()
screen.mainloop()
```

OUTPUT

The following screenshots show the Rocket Space Shooter game during gameplay and the Game Over state.

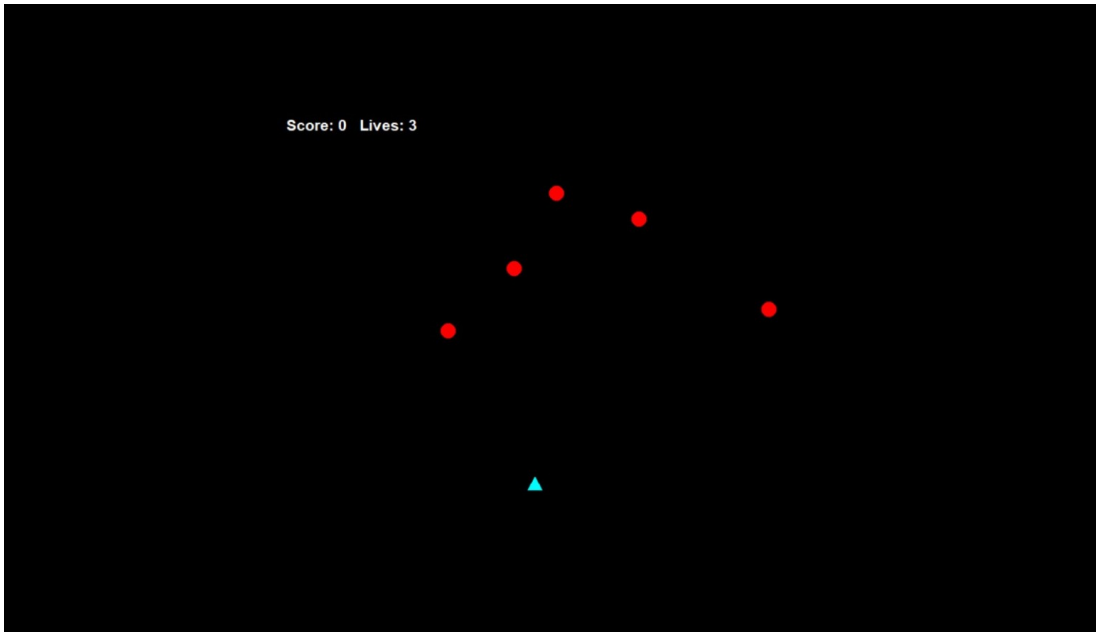


Figure 1: Rocket Space Shooter during gameplay.

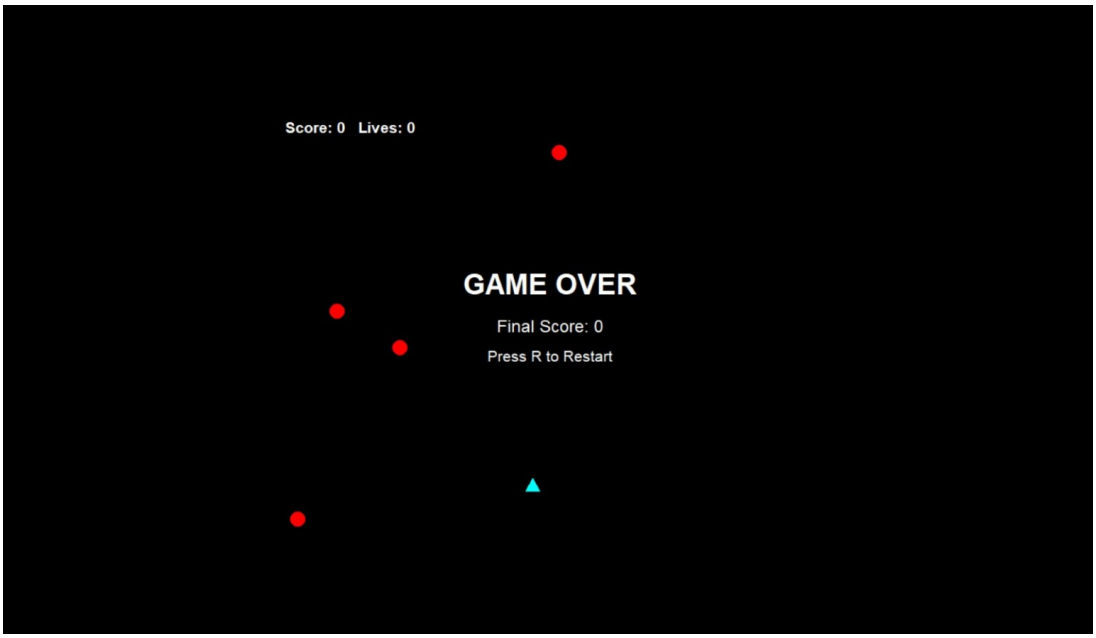


Figure 2: Game Over screen showing the final score and restart instruction.

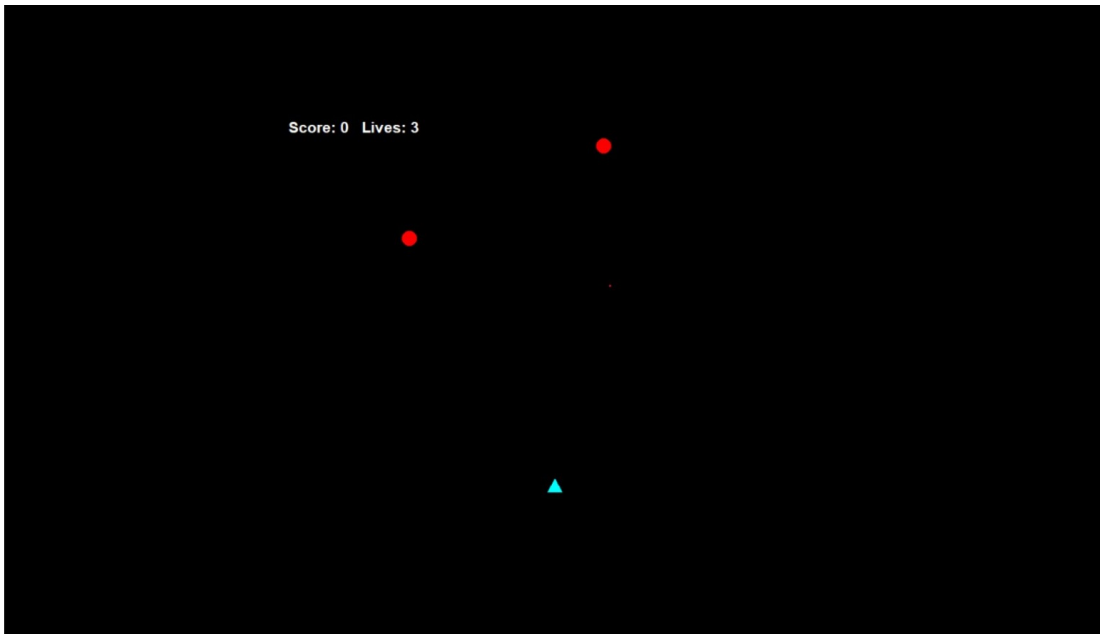


Figure 3: Gameplay showing the rocket and incoming enemies.

RESULT

Thus, an interactive Rocket Space Shooter game was developed using Python Turtle with graphical game design, keyboard-based movement, bullet shooting, random enemy generation, automatic score calculation, collision detection, lives management, game-over handling, and restart functionality. The program demonstrates the use of Python libraries, functions, loops, conditional statements, lists, random number generation, keyboard event handling, and basic game development concepts.